

Classifying Adolescent Excessive Alcohol Drinkers from fMRI Data

UE24CS352A Machine Learning · Mini-Project #35 | Harshitha P (PES1UG24CS186) · Gautam Prasanth Nair (PES1UG24CS169)

1. Problem Statement

Heavy drinking during adolescence is known to alter brain *structure*, but whether it leaves a detectable signature in brain *function* is still open. The task is a binary classifier that predicts whether a 16–19-year-old is a **heavy drinker** from their resting-state fMRI (BOLD signal) and basic demographics. We replicate the models of the reference study (Kim, Liu & Noh, Stanford CS229: logistic regression, SVMs, CNN/RNN). We then test the ideas it lists as future work: richer fMRI features, demographics inside the models and more rigorous validation.

2. Dataset

The reference study uses 715 NCANDA subjects. Each subject has $T = 269$ volumes ($TR = 2.2$ s), parcellated into $N = 25$ ICA components or $N = 100$ Craddock regions, plus age, sex and scanner type. 122 subjects (17%) are heavy drinkers. NCANDA is released only under a data-use agreement, so we wrote a **documented simulator** with exactly this structure ($m \times T \times N$ tensors, same class ratio, demographics). Each region follows one of five networks (DMN, salience, executive, visual, somatomotor) driven by slow AR(1) latent signals. On top of that the simulator adds a global signal, scanner-dependent noise, motion spikes and a raw scanner baseline. Heavy drinkers differ by small, explicit effects: weaker DMN/ECN coupling, stronger salience–DMN coupling, and higher prevalence with age. The code loads real data in the same .npz format without changes. **Our numbers describe how the methods behave on NCANDA-shaped data. They are not clinical findings.**

3. Approach

- **Pre-processing (as in the paper):** z-score each region's time series; undersample the majority class to a 50/50 set of 244 subjects; hold out a stratified 5% **dev set** (13 subjects) that never enters cross-validation; run 10-fold stratified CV on the remaining 231. We repeat the whole procedure over 5 random undersamplings and report mean $\pm$ sd.
- **Features:** the paper's derived dynamic range $x(N) = \max(x_N) - \min(x_N)$; our **functional connectivity** (Fisher-z Pearson correlation of every region pair: 300 features for ICA-25, 4,950 for Craddock-100); demographics.
- **Models:** logistic regression with **Newton's method** (from scratch, $\theta := \theta - H^{-1}\nabla l(\theta)$, tol 1e-5); SVMs (linear, poly-2, sigmoid, RBF) with C/γ tuned by inner CV; PyTorch RNN, RNN+NN, CNN+NN and CNN→LSTM→Dense (window 5, stride 1, 5 filters, RNN dim 5), trained with batch BCE and the paper's three dev-set stopping rules. Extensions: L2-regularised LR on connectivity, a **late-fusion** model (averaged log-odds of a demographics LR and a connectivity LR), and a CNN+RNN that takes demographics as a second input.
- **Validation beyond the paper:** ROC-AUC alongside accuracy/F1; age ablations; a full-cohort protocol (all 715 subjects, class-weighted, stratified 10-fold, balanced accuracy); a 200-permutation label test.

4. Implementation

Python package `fmri_alcohol`: `data` (simulator + loader), `preprocess`, `features`, `models/` (Newton LR, SVM/LR factories, PyTorch networks), `evaluate` (protocols, metrics) and `plots`. Scripts: `generate_data.py`, `run_experiments.py` (writes every table and figure to `results/`) and `demo.py` (live prediction on a held-out subject, with a what-if age option). 15 pytest unit tests cover the pipeline. They check, among other things, that no subject is shared between train/test/dev splits and that our Newton LR matches scikit-learn's coefficients to 1e-4. All tuning happens inside training folds, so test folds are never used to choose hyper-parameters.

5. Results

Model	Features	ICA-25 Acc	F1	AUC	Craddock Acc	AUC
LR (paper baseline)	range + demo	0.62±0.11	0.62	0.66	0.52	0.52
LR – age	range + sex, scanner	0.55±0.11	0.54	0.54	0.48	0.47
LR fMRI only	range	0.52±0.10	0.52	0.52	0.50	0.49
SVM-Linear	range + demo	0.63±0.11	0.63	0.67	0.61	0.63
SVM-RBF	range + demo	0.63±0.10	0.64	0.68	0.60	0.63
RNN only	raw BOLD	0.51±0.13	0.49	0.54	0.43	0.41
CNN+RNN+Dense	raw BOLD	0.48±0.02	0.25	0.45	0.48	0.54
CNN+RNN+demo	raw BOLD + demo	0.46±0.07	0.38	0.44	0.51	0.53
CNN+RNN, no stop rules	raw BOLD	0.49±0.13	0.47	0.49	0.51	0.52
Age only	age	0.66±0.10	0.67	0.70	0.66	0.70
FC-LR	conn + demo	0.64±0.10	0.64	0.71	0.64	0.70
FC-LR – age	conn + sex, scanner	0.61±0.11	0.61	0.67	0.64	0.70
FC-LR fMRI only	conn	0.61±0.11	0.62	0.67	0.64	0.70
FC + Demo fusion	conn $\oplus$ demo	0.67±0.10	0.68	0.75	0.70	0.76

Table 1. 10-fold CV on the balanced set (mean over folds $\times$ 5 undersamplings; deep models 1×10 folds). Rows: paper models / deep models / our extensions. Highlighted: best model on both parcellations.

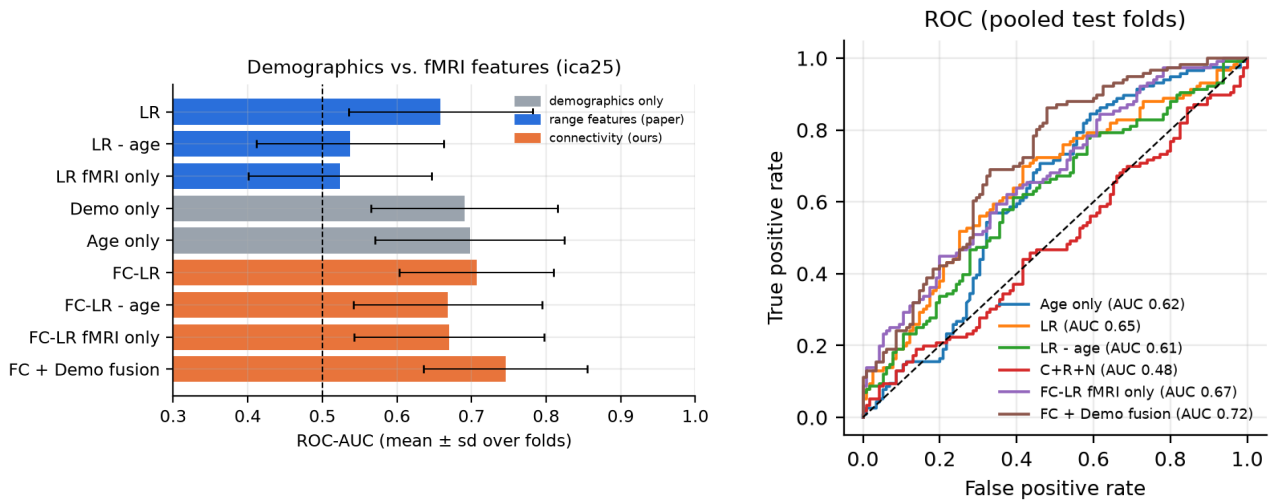


Figure 1. Left: demographics vs. fMRI features (mean per-fold AUC). Right: ROC curves pooled over the test folds of one repeat, so AUCs differ slightly from Table 1 (ICA-25).

- **Replication.** As in the paper, LR on derived range features + demographics is a reasonable baseline (acc 0.62), but removing age drops it to 0.55, and range features alone give 0.52 (chance). Age alone reaches 0.66, so the baseline's performance comes almost entirely from demographics. SVMs perform about the same (RBF 0.63, linear 0.63).
- **Deep learning.** All deep models stay at chance (best: R, 0.51). With the paper's stopping rules, the 13-subject dev set ends training after about 10 epochs on noise, and the models often predict one class (C+R+N recall 0.28). With the rules turned off, C+R+N reaches train accuracy 0.84 but test 0.49: it overfits. Adding demographics (0.46) does not help. About 200 subjects are too few for these networks, which matches the reference study.
- **Connectivity carries signal that range features miss.** Without age, FC-LR keeps 0.61 (range features: 0.55). fMRI-only FC gives 0.61 on ICA-25 and 0.64 on Craddock-100 (permutation $p = 0.005$ and 0.010). Late fusion is the best model: acc 0.67 / AUC 0.75 (ICA-25) and **0.70 / 0.76** (Craddock-100). Regularisation decides the parcellation result. Unregularised Newton LR on Craddock's 103 range features overfits (train 0.88, test 0.52). L2-penalised FC copes with 4,950 features, but its shared penalty then ignores the demographic columns (FC-LR $0.64 \approx$ FC-LR – age 0.64). Fusion fixes this by giving demographics their own model.
- **Full cohort.** Using all 715 subjects with class weights instead of discarding 471 controls, FC-LR gets AUC 0.73 vs. 0.68 for demographics only (balanced accuracy 0.65). Undersampling throws away useful data.

6. Conclusions

We rebuilt the reference study end to end and reproduced its main result: complex deep models underperform a simple logistic regression, and that regression depends on age rather than on brain activity. The fix is better *features*, not deeper models. Between-region functional connectivity, a strong L2 penalty, and a separate model for demographics recover an fMRI signal that stays significant after removing age and passes a permutation test. Methodologically, a 13-subject dev set is too small to drive early stopping; repeated CV with AUC and all subjects is more reliable than a single undersampled split. Next steps: run the same pipeline on real NCANDA data (it loads as-is), regress out age/site confounds explicitly, try graph neural networks on the connectivity matrix, and use transfer learning from large resting-state cohorts to overcome the small sample size.

Code: private GitHub repository `fmri-alcohol-classifier` (README has setup and run instructions). **Ref.:** Kim, Liu, Noh, *Classifying Adolescent Excessive Alcohol Drinkers from fMRI Data*, CS229 Stanford; NCANDA (ncanda.org); Craddock et al., *Hum. Brain Mapp.* 2012.